

## SAFETY DATA SHEET

Revision Date 01-April-2024

Revision Number 4

### 1. Identification

**Product Name** Polydimethylsiloxane (silicone) emulsion

**Cat No. :** 45088

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### Importer/Distributor

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### Emergency Telephone Number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Not classified under the Hazardous Products Regulations (SOR/2015-17)

Based on available data, the classification criteria are not met

#### Label Elements

None required

### 3. Composition/Information on Ingredients

| Component              | CAS-No     | Weight % |
|------------------------|------------|----------|
| Water                  | 7732-18-5  | 45.00    |
| Poly(dimethylsiloxane) | 63148-62-9 | 45       |
| Sorbitan monolaurate   | 1338-39-2  | 5        |

|                                        |           |      |
|----------------------------------------|-----------|------|
| Polyoxyethylene 20 sorbitan monooleate | 9005-65-6 | 5.00 |
|----------------------------------------|-----------|------|

#### 4. First-aid measures

|                                                               |                                                                                                                         |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                                           | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Inhalation</b>                                             | Remove to fresh air. Get medical attention immediately if symptoms occur.                                               |
| <b>Ingestion</b>                                              | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                   |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | None reasonably foreseeable.<br>Treat symptomatically                                                                   |

#### 5. Fire-fighting measures

|                                         |                                                                                                                                                                      |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Powder. Water spray. In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                             |
| <b>Flash Point</b>                      | No information available                                                                                                                                             |
| <b>Method -</b>                         | No information available                                                                                                                                             |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                                             |
| <b>Explosion Limits</b>                 |                                                                                                                                                                      |
| <b>Upper</b>                            | No data available                                                                                                                                                    |
| <b>Lower</b>                            | No data available                                                                                                                                                    |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                             |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                             |

#### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Silicon dioxide.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

**Health**  
0

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
-

#### 6. Accidental release measures

|                                  |                                                                             |
|----------------------------------|-----------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment as required. |
| <b>Environmental Precautions</b> | Should not be released into the environment.                                |

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal.

#### 7. Handling and storage

|                 |                                                                                        |
|-----------------|----------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid |
|-----------------|----------------------------------------------------------------------------------------|

contact with skin, eyes or clothing. Avoid ingestion and inhalation.

**Storage.**

Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Acids. Strong bases. Oxidizing agent.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

None under normal use conditions.

**Personal protective equipment****Eye Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                  |                          |
|----------------------------------|--------------------------|
| Physical State                   | Liquid                   |
| Appearance                       | No information available |
| Odor                             | No information available |
| Odor Threshold                   | No information available |
| pH                               | Not applicable           |
| Melting Point/Range              | No data available        |
| Boiling Point/Range              | No information available |
| Flash Point                      | No information available |
| Evaporation Rate                 | No information available |
| Flammability (solid,gas)         | Not applicable           |
| Flammability or explosive limits |                          |

|                                        |                          |
|----------------------------------------|--------------------------|
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | <=1100 hPa @ 50 °C       |
| Vapor Density                          | No information available |
| Specific Gravity                       | No information available |
| Solubility                             | Dispersible in water     |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |

## 10. Stability and reactivity

|                                  |                                                                          |
|----------------------------------|--------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                               |
| Stability                        | Stable under normal conditions.                                          |
| Conditions to Avoid              | Incompatible products.                                                   |
| Incompatible Materials           | Acids, Strong bases, Oxidizing agent                                     |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Silicon dioxide |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                 |
| Hazardous Reactions              | None under normal processing.                                            |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Oral LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

##### Dermal LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

##### Vapor LC50

Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

#### Component Information

| Component                              | LD50 Oral                  | LD50 Dermal | LC50 Inhalation                           |
|----------------------------------------|----------------------------|-------------|-------------------------------------------|
| Water                                  | -                          | -           | -                                         |
| Poly(dimethylsiloxane)                 | LD50 > 24 g/kg ( Rat )     | Not listed  | Not listed                                |
| Sorbitan monolaurate                   | LD50 = 33600 mg/kg ( Rat ) | Not listed  | LC50 > 5000 mg/m <sup>3</sup> ( Rat ) 4 h |
| Polyoxyethylene 20 sorbitan monooleate | LD50 = 34500 µL/kg ( Rat ) | Not listed  | Not listed                                |

**Toxicologically Synergistic** No information available

#### Products

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                              | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------------------|------------|------------|------------|------------|------------|------------|
| Water                                  | 7732-18-5  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Poly(dimethylsiloxane)                 | 63148-62-9 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Sorbitan monolaurate                   | 1338-39-2  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Polyoxyethylene 20 sorbitan monooleate | 9005-65-6  | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

|                                                   |                                                                |
|---------------------------------------------------|----------------------------------------------------------------|
| <b>Reproductive Effects</b>                       | No information available.                                      |
| <b>Developmental Effects</b>                      | No information available.                                      |
| <b>Teratogenicity</b>                             | No information available.                                      |
| <b>STOT - single exposure</b>                     | None known                                                     |
| <b>STOT - repeated exposure</b>                   | None known                                                     |
| <b>Aspiration hazard</b>                          | No information available                                       |
| <b>Symptoms / effects, both acute and delayed</b> | No information available                                       |
| <b>Endocrine Disruptor Information</b>            | No information available                                       |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated. |

## 12. Ecological information

### Ecotoxicity

| Component                              | Freshwater Algae | Freshwater Fish                    | Microtox   | Water Flea |
|----------------------------------------|------------------|------------------------------------|------------|------------|
| Polyoxyethylene 20 sorbitan monooleate | Not listed       | LC50: 471 mg/L/96h (Rainbow trout) | Not listed | Not listed |

**Persistence and Degradability** Immiscible with water

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

|                 |               |
|-----------------|---------------|
| <b>DOT</b>      | Not regulated |
| <b>TDG</b>      | Not regulated |
| <b>IATA</b>     | Not regulated |
| <b>IMDG/IMO</b> | Not regulated |

## 15. Regulatory information

**All of the components in the product are on the following inventory lists:** China X = listed Australia U.S.A. (TSCA) Canada (DSL/NDL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (KECL) China (IECSC) Japan (ENCS) Philippines (PICCS)

### International Inventories

| Component                              | CAS-No     | DSL | NDL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP       |
|----------------------------------------|------------|-----|-----|------|-----------------------------------------------|-----------|--------|-----------|
| Water                                  | 7732-18-5  | X   | -   | X    | ACTIVE                                        | 231-791-2 | -      | -         |
| Poly(dimethylsiloxane)                 | 63148-62-9 | X   | -   | X    | ACTIVE                                        | -         | -      | -         |
| Sorbitan monolaurate                   | 1338-39-2  | X   | -   | X    | ACTIVE                                        | 215-663-3 | -      | -         |
| Polyoxyethylene 20 sorbitan monooleate | 9005-65-6  | X   | -   | X    | ACTIVE                                        | -         | -      | 500-019-9 |

| Component | CAS-No | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------|--------|-------|------|------|------|------|------|-------|-------|
|-----------|--------|-------|------|------|------|------|------|-------|-------|

|                                        |            |   |          |   |   |   |   |   |   |
|----------------------------------------|------------|---|----------|---|---|---|---|---|---|
| Water                                  | 7732-18-5  | X | KE-35400 | X | - | X | X | X | X |
| Poly(dimethylsiloxane)                 | 63148-62-9 | X | KE-31068 | X | - | X | X | X | X |
| Sorbitan monolaurate                   | 1338-39-2  | X | KE-31687 | X | X | X | X | X | X |
| Polyoxyethylene 20 sorbitan monooleate | 9005-65-6  | X | KE-25511 | X | X | X | X | X | X |

**Legend:**

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations**

**Authorisation/Restrictions according to EU REACH** Not applicable

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                              | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Water                                  | 7732-18-5  | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Poly(dimethylsiloxane)                 | 63148-62-9 | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Sorbitan monolaurate                   | 1338-39-2  | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Polyoxyethylene 20 sorbitan monooleate | 9005-65-6  | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                              | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water                                  | 7732-18-5  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Poly(dimethylsiloxane)                 | 63148-62-9 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Sorbitan monolaurate                   | 1338-39-2  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Polyoxyethylene 20 sorbitan monooleate | 9005-65-6  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

Product Safety Department  
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**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

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**End of SDS**